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Conscientiousness, prior experience, achievement emotions and academic procrastination in online learning environments

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Abstract

Academic procrastination refers to individuals' unnecessary postponement of their coursework and is harmful for academic performance. When situated in self-placed and remote learning environments, students' tendency to procrastinate increases. Therefore, understanding why students procrastinate and identifying who is more likely to delay unnecessarily in online learning environments is an important area to study. The goal of this study was to respond to this call by examining the structural relations between conscientiousness, prior online learning experience, achievement emotions and academic procrastination in online learning environments using structural equation modelling. In particular, two main facets of conscientiousness—proactive and inhibitive—were examined in order to understand which facet was more responsible for procrastination, how each facet was related to procrastination and to determine which facet of conscientiousness should be the primary target for intervention in future experimental research. A total of 746 students from 49 secondary and postsecondary schools participated in the current study. The results showed that the proactive aspect of conscientiousness was negatively related to academic procrastination through the pathway of enjoyment. The inhibitive aspect of conscientiousness was negatively related to academic procrastination through the pathways of negative emotions.

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Although prior online learning experience did not have a significant and direct association with academic procrastination, it was indirectly related to students' tendency to put off coursework through the pathways of enjoyment and negative emotions. Of all the paths, negative emotions had the strongest associations with academic procrastination in online learning environments. The findings of indirect effects suggest that students with low conscientiousness, the inhibitive aspect in particular, and few prior online learning experiences are more likely to procrastinate in online classes possibly owing to the suboptimal emotional experiences aroused during their learning processes.

KEYWORDS
academic procrastination, achievement emotions, conscientiousness, online learning, prior online learning experience

Practitioner notes

What is already known about this topic

- Conscientiousness has a direct association with academic procrastination in in-person classes.
- Prior online learning experience has direct associations with students' adaptive regulatory processes in online classes.
- Achievement emotion is an integral part of online learning.

What this paper adds

- Conscientiousness has a direct association with academic procrastination in online learning environments.
- Prior online learning experience has no direct associations with academic procrastination in online learning environments.
- Conscientiousness indirectly relates to academic procrastination through achievement emotions in online learning environments.
- Prior online learning experience indirectly relates to academic procrastination through achievement emotions in online learning environments.

Implications for practice and/or policy

- Instructors are suggested to pay attention to students with low conscientiousness and few prior online learning experiences since they are more likely to procrastinate in online learning environments.
- Students' tendency to procrastinate in online learning environments is strongly associated with their emotional experience.

INTRODUCTION

Online learning is appealing to many students thanks to its convenience and flexibility (Lee et al., 2017). Having the autonomy to choose when and where to study allows students to

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work while continuing their education (Harris & Martin, 2012; Jaggars, 2014). However, this high level of freedom is accompanied by a need for students to be proactive in managing their own learning. It is not surprising that one of the main challenges for students to successfully learn in self-paced and remote learning environments is procrastination (Hendrickson et al., 2021). Around 62.4% of college students reported a greater tendency to procrastinate when transitioning from face-to-face classes to online learning environments (Usher et al., 2021). Procrastination reflects individuals' voluntary decision to postpone their course of action despite expecting that such delay is not beneficial for their utilities (Steel, 2007). Meta-analytic studies have consistently shown that procrastination is detrimental to students' well-being and academic performance (Kim & Seo, 2015; Steel, 2007; van Eerde, 2003). Given that students have a greater tendency to procrastinate in self-paced and remote settings, and procrastination is harmful for learning, studies on procrastination in online learning environments are, hence, warranted.

The existing literature has provided preliminary and important insights into the occurrence of procrastination in online learning environments. Many studies have found that motivational beliefs and learning strategy use are the major factors that are associated with students' unnecessary postponement of coursework in online classes (Dunn, 2014; Ma et al., 2022; Pelikan et al., 2021). However, the recent development in the theory has offered a new perspective on the incidence of procrastination. That is, negative affective states precede procrastination. When individuals experience negative emotions during the goal-directed process, they are more likely to succumb to procrastination to avoid task engagement and to seek immediate mood repair (Baumeister et al., 2007; Sirois, 2016; Sirois & Pychyl, 2013). Therefore, apart from motivational beliefs and learning strategy use, emotion is proposed as another important activator of procrastination. Although emotion has been reported as an integral part of online learning experience (Baltà-Salvador et al., 2021; Cleveland-Innes & Campbell, 2012; Sim et al., 2021), the link between emotion and procrastination in online learning environments remains understudied. On the other hand, in theory, personal characteristics have been proposed as important antecedents of students' emotional experiences and their tendency to procrastinate (Pekrun, 2006; Schraw et al., 2007). Specifically, conscientiousness and prior online learning experience have been shown to be important personal characteristics that are critical for student success in online learning environments (Abdous, 2019; Abe, 2020). However, studies on how conscientiousness and prior online learning experience are related to students' emotional experiences and academic procrastination in online settings remain scarce. The goal of the current study was to address these gaps with an aim to identify who is likely to procrastinate and reveal why these students are prone to delay unnecessarily in online learning environments. This inquiry will not only advance the theory on procrastination but also offer practical implications for instructors who teach online.

LITERATURE REVIEW

Academic procrastination in online learning environments

Academic procrastination refers to students' unnecessary postponement of schoolwork despite their awareness of negative consequences that will result from such delay (Cheng & Xie, 2021). Academic procrastination is detrimental to all kinds of academic achievement such as GPA, assignment grade, course grade and exam scores as meta-analyses have shown (Kim & Seo, 2015; Steel, 2007; van Eerde, 2003). Procrastination has been conceptualized as self-regulation failure (Steel, 2007). That is, academic procrastination stems from students' inability to plan, monitor and control their learning. Given the unique nature of online learning environments with a larger degree of flexibility and freedom accompa-

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nied by distractions and fewer regulatory cues from instructors and peers (Jaggars, 2014; Kalman et al., 2020), procrastination becomes a common challenge for students who transition from traditional classrooms to online courses (Usher et al., 2021). Recently, scholars have proposed that individuals' unnecessary delay does not merely result from a lack of goal setting and time management; rather, procrastination is the outcome of emotions (Sirois, 2016; Sirois & Pychyl, 2013). When individuals experience negative feelings during the goal-directed process, they resort to procrastination as a way to avoid task engagement and "a strategy to avoid this negative mood" (Sirois & Pychyl, 2013, p. 117). However, as research has shown, procrastination is not beneficial for task performance (Pinxten et al., 2019). Therefore, procrastination can be considered a counterproductive strategy in response to negative emotions caused by aversive tasks. Many studies have shown that negative affective states tend to prompt individuals to engage in maladaptive self-regulatory processes or the so-called "mis-regulation" such as procrastination (Balkis & Duru, 2016; Baumeister et al., 2007). In the current study, we consider academic procrastination a type of self-regulation that is not beneficial for goal attainment and the primary outcome variable of interest. As suggested by the existing literature, individuals' emotions play a major role in contributing to the occurrence of procrastination. In the next session, we will discuss the relation between emotions and procrastination in academic settings.

Achievement emotions and academic procrastination

According to the control-value theory, achievement emotions are defined as "multi-component, coordinated processes of psychological subsystems including affective, cognitive, motivational, expressive, and peripheral physiological processes" related to activities or outcomes in academic settings (Pekrun, 2006, p. 316). Achievement emotions are multifaceted and multidimensional and generally can be categorized according to object focus (activity and outcome), valence (positive and negative) and activation degree (activating and deactivating). Objective focus pertains to whether emotions are directly tied to ongoing academic activities or the outcomes of these activities (Pekrun, 2006). Valence pertains to the hedonic tone of emotions (Linnebrink, 2007). Activation pertains to the degree of "arousal, mobilization, and energy" associated with emotions (Linnebrink, 2007, p. 108). Therefore, achievement emotions can be organized into a 3-dimensional framework (Pekrun et al., 2007). For instance, enjoyment is considered a positive activating activity emotion, while anger and frustration are negative activating activity emotions. Contentment and relief are positive deactivating outcome emotions, while sadness and hopelessness are negative deactivating outcome emotions. It is hypothesized that positive emotional experiences are conducive to cognitive functioning and flexible and creative ways of approaching academic tasks, which facilitate adaptive self-regulation and hinder maladaptive self-regulation (Pekrun et al., 2002, 2007). Therefore, positive emotions are assumed to be negatively related to academic procrastination, while negative emotions are assumed to be positively associated with it.

Emotion is an integral part of online learning. Students experience various emotions when learning in a self-paced and remote setting such as enjoyment (Cleveland-Innes & Campbell, 2012), boredom (Baltà-Salvador et al., 2021), anger and hopelessness (Stephan et al., 2019), and anxiety (Zembylas, 2008). Although research in face-to-face settings have shown that academic procrastination was significantly and negatively correlated with enjoyment and positively correlated with boredom, anger, hopelessness and anxiety (Rahimi, 2019), how these emotions relate to academic procrastination in online learning environments has not been widely examined. The existing literature has focused on the relation between achievement emotions (ie, enjoyment, boredom, anger, hopelessness and anxiety) and students' adaptive self-regulation (Artino & Jones, 2012; Kim et al., 2014; You &

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Kang, 2014). Recently, a few studies have started to explore the link between achievement emotions and students' maladaptive self-regulation, that is, academic procrastination, in remote learning settings. For instance, Henderikx et al. (2019) found that boredom was significantly and positively correlated with academic procrastination. However, no significant correlation was found for enjoyment among MOOC learners. Rahimi and Vallerand (2021) also found that negative emotions were significantly and positively correlated with academic procrastination, while no significant correlations were found for positive emotions among undergraduate students. These studies provided preliminary insights into the associations between achievement emotions and academic procrastination in online learning environments. However, how other emotions such as anger, anxiety, hopelessness and boredom are related to online procrastination has not been examined. In addition, the fact that enjoyment was not significantly correlated with academic procrastination as found by Henderikx et al. (2019) and Rahimi and Vallerand (2021) contradicts the theory and the findings reported in face-to-face settings (Rahimi, 2019). Therefore, there is a need to expand the scope of past work by revisiting the association between online academic procrastination and multiple aspects of achievement emotions. In the current study, we consider achievement emotions as the antecedents of academic procrastination. Based on the review of the relevant literature, we tested the following hypothesis:

Hypothesis 1. *Positive emotions do not relate to academic procrastination, while negative emotions positively relate to academic procrastination in online learning environments.*

Conscientiousness and academic procrastination

Conscientiousness is defined as an individual's long-lasting patterns of thoughts, feelings and behaviours that show the tendency to be persistent, self-controlled, organized and hard-working (Roberts et al., 2014). In theory, conscientiousness is conceptualized as a multidimensional construct, consisting of two major aspects (Roberts et al., 2005). The proactive aspect of conscientiousness encompasses concepts related to "the need for achievement and commitment to work" (Costa et al., 1991, p. 889). The inhibitive aspect of conscientiousness encompasses concepts related to impulse control (Costantini & Perugini, 2016) and "moral scrupulousness and cautiousness" (Costa et al., 1991, p. 889). Since both aspects represent the same valence (Roberts et al., 2005), conscientiousness normally is treated as a unidimensional construct in empirical studies (Oshio et al., 2018). Highly conscientious individuals have a cognitive, metacognitive, affective and behavioural propensity to work hard, perceive high control over their action and outcomes, and be strategic about completing tasks (Mirhashemi & Goodarzi, 2014). In addition, they rarely act on an impulse, do not get distracted easily and persist in the face of challenges (Roberts et al., 2014). These characteristics suggest that conscientiousness is assumed to be positively related to students' adaptive self-regulatory processes and negatively related to maladaptive regulatory processes. Empirical findings support this assumption as meta-analyses have consistently shown that conscientiousness is significantly and negatively correlated with procrastination ($\bar{r}$ = −0.62 and −0.63; see Steel, 2007; van Eerde, 2003). That is, conscientious students are associated with a lower likelihood of engaging in unnecessary delay. However, this relation has not been examined much in online learning environments. One study by Cheng & Xie (2021) showed that conscientiousness was significantly and negatively correlated with academic procrastination in online courses. However, the magnitude was much smaller (r = −0.28). In the current study, we consider conscientiousness as an antecedent of academic procrastination. Based on the review of the relevant literature, we tested the following hypothesis:

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Hypothesis 2. *Conscientiousness negatively relates to academic procrastination in online learning environments.*

Conscientiousness and achievement emotions

Apart from being an immediate precursor of academic procrastination, conscientiousness is also assumed to be an important antecedent of affective states as suggested by the control-value theory of achievement emotions (Pekrun, 2006; Pekrun et al., 2002). That is, an individual's propensity to perceive high control over their action and outcomes should lead to the arousal of positive emotions and the suppression of negative emotions (Pekrun et al., 2007). Empirical findings also support this assumption. For instance, Sorić et al. (2013) found that conscientiousness was significantly and negatively correlated with unhappiness, anger, anxiety and humiliation among high school students. A study by de la Fuente et al. (2020) showed that conscientiousness was significantly and positively correlated with enjoyment, hope and pride and negatively correlated with boredom, anger, anxiety, shame and hopelessness among college students.

Instead of examining conscientiousness as a global construct in relation to emotions, some scholars took a step further to investigate how the subdomains of conscientiousness—proactive and inhibitive components—relate to emotions. For instance, Watson and Clark (1992) and Fayard (2012) found that both proactive and inhibitive conscientiousness were significant and positively correlated with positive emotions with the magnitude of the association greater for proactive conscientiousness. Focusing on pride, Carver et al. (2010) reported that both proactive and inhibitive conscientiousness were significant and positively correlated with adaptive pride (ie, authentic pride) with the magnitude of the association greater for proactive conscientiousness. In addition, both proactive and inhibitive conscientiousness were significant and positively correlated with maladaptive pride (ie, hubristic pride) with the magnitude of the association greater for inhibitive conscientiousness. Using network analysis, Costantini and Perugini (2016) found that proactive conscientiousness was closely associated with adaptive pride (ie, authentic pride), while inhibitive conscientiousness was closely associated with low maladaptive pride (ie, hubristic pride). Compared to positive emotions, few studies have examined facets of conscientiousness and negative emotions. One study showed that both proactive and inhibitive conscientiousness were significantly and negatively correlated with guilt. The magnitude of the association was greater for inhibitive conscientiousness (Fayard et al., 2012).

Taken together, the existing literature has provided preliminary insights into the relations between facets of conscientiousness and emotions. The findings suggest that proactive conscientiousness is more closely related to positive emotions, while inhibitive conscientiousness is more closely related to negative emotions. However, prior studies are normally situated in daily life settings, which can be different from classroom environments. Besides, few studies, thus far, have examined how proactive and inhibitive conscientiousness are related to diverse emotions in achievement-related contexts. Whether the relational patterns observed in the literature are identical in online learning environments remains a question to be answered. In the current study, we consider conscientiousness as an antecedent of achievement emotions. Based on the review of the relevant literature, we tested the following hypothesis:

Hypothesis 3. *Conscientiousness positively relates to positive emotions and negatively relates to negative emotions in online learning environments.*

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Prior online learning experience and academic procrastination

According to the general learning model proposed by Pintrich and Zusho (2007), personal characteristics such as demographics, personality and other relatively stable attributes students bring with them to classrooms act as important antecedents of self-regulation. One of the personal characteristics that is critical for student success in online learning environments is prior online learning experience (Abdous, 2019). Prior online learning experience can be defined as students' perceptions of how experienced they are with self-paced, online learning prior to enrolling in the current online courses (Artino, 2008b). Experienced online learners are equipped with more system knowledge and adaptive motivational beliefs to use computer-based applications tools and navigate web-based distance learning systems (Artino, 2007; Hannafin et al., 2003; Jan, 2015), which serve as activators of the benign cycle of self-regulatory processes (Panadero, 2017). Thus far, the existing literature has mainly focused on the relation between prior online learning experience and students' adaptive self-regulation (Li, 2019; Oyelere et al., 2021; Wang et al., 2013). These prior studies have consistently shown that prior online learning experience is significantly and positively related to students' adaptive self-regulation such as the use of cognitive and metacognitive strategies, goal setting and environmental restructuring in online learning environments. However, few studies, thus far, have examined the relation between prior online learning experience and students' maladaptive self-regulation such as academic procrastination in online learning environments. Although academic procrastination and adaptive self-regulation differ in their valences (ie, maladaptive vs. adaptive), both constructs are conceptually related and can be subsumed under the broader category of self-regulation (Baumeister et al., 1994). Given that personal characteristics are hypothesized to be important antecedents of self-regulation (Pintrich & Zusho, 2007), one logical extension of the previous findings is that prior online learning experience is expected to be negatively related to students' maladaptive self-regulation, that is, academic procrastination in online learning environments. The more experience students have in learning in a remote setting, the less likely they will procrastinate in such environment. In the current study, we consider prior online learning experience as an antecedent of academic procrastination. Based on the review of the relevant literature, we tested the following hypothesis:

Hypothesis 4. *Prior online learning experience negatively relates to academic procrastination in online learning environments.*

Prior online learning experience and achievement emotions

As suggested by the principles of relative universality (Pekrun, 2018), the mechanisms underlying achievement emotion are universal across individuals. However, the intensity, frequency and patterns of emotions could differ by personal characteristics such as prior online learning experience. A few studies have been conducted to test this hypothesis. For instance, Artino (2008a) found that prior online learning experience was significantly and negatively correlated with boredom and frustration in a self-paced online course. Abdous (2019) also found that prior online learning experience significantly and negatively related to online students' feelings of anxiety. Thus far, research on the link between prior online learning experience and students' diverse emotional experiences in online learning environments is still limited. In the current study, we consider prior online learning experience as an antecedent of achievement emotions. Based on the review of the relevant literature, we tested the following hypothesis:

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Hypothesis 5. *Prior online learning experience positively relates to positive emotions and negatively relates to negative emotions in online learning environments.*

Indirect relations between conscientiousness, prior online learning experience, achievement emotions and academic procrastination

According to the control-value theory, achievement emotions mediate the relations between personal characteristics and students' self-regulation (Pekrun, 2006). This hypothesized relation has also been verified by studies situated in face-to-face learning environments (Chou, 2021). However, studies on how personal characteristics are indirectly related to students' self-regulation such as academic procrastination through the mechanism of achievement emotions in online learning environments remain scarce. As shown in previous sections, conscientiousness and prior online learning experience are significantly associated with achievement emotions (Abdous, 2019; Artino, 2008a; de la Fuente et al., 2020; Sorić et al., 2013). Achievement emotions are also significantly associated with academic procrastination (Henderikx et al., 2019; Rahimi & Vallerand, 2021). Therefore, apart from the direct effects of conscientiousness and prior online learning experience on academic procrastination, it is expected that these two personal characteristics are indirectly related to academic procrastination through the mechanisms of achievement emotions in online learning environments. In the current study, we consider achievement emotions as the mediators through which conscientiousness and prior online learning experience are related to academic procrastination in online learning environments. Based on the review of the relevant literature, we tested the following hypothesis:

Hypothesis 6. *Conscientiousness indirectly relates to academic procrastination through the pathways of achievement emotions in online learning environments.*

Hypothesis 7. *Prior online learning experience indirectly relates to academic procrastination through the pathways of achievement emotions in online learning environments.*

Significance, research questions and hypotheses

In summary, online learning has become an integral part of formal education since the global pandemic of COVID-19. Although online learning features convenience and flexibility, students' tendency to procrastinate in a self-paced remote learning environment also increases. Therefore, examining who is more likely to procrastinate and why procrastination occurs among these students in online learning environments is important for theory building and online instruction. Although the hypothesized relations between conscientiousness, prior online learning environments, achievement emotions and academic procrastination have strong theoretical bases, empirical research on these relations in the context of online learning environments remains scarce. The goal of the current study was to address the gaps in the existing literature.

METHOD

Research setting, procedure and participants

The sampling frame of the current study was 1150 students from 49 secondary and post-secondary schools (grade 10–16) in Taiwan. We received a total of 791 responses. After

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removing duplicates and participants who did not identify themselves as secondary or post-secondary students, there were 746 valid cases. Among these participants, 397 (53.22%) were females and 349 (46.78%) were males with an average age of 18.02 (*SD* = 2.74). In terms of grade levels, 507 were enrolled in secondary schools (67.96%) and 239 were in postsecondary schools (32.04%).

In response to the COVID-19 pandemic, it was ordered that starting May 19 in 2021, all physical classes across all levels of education be suspended and transitioned to online learning. The context is ideal for the present study in that in-person instruction is the main modality in the partner schools. Rarely was there a situation where online learning was implemented across all levels of education in Taiwan at the same time. The school lockdown allowed us to recruit a large sample of students who were learning online at the same time from these schools. Also, students were placed in a new and technology-rich learning environment, which constituted a natural setting to examine how their personal characteristics were related to emotional experiences and their tendency to procrastinate in online learning environments.

About 2 weeks after the school lockdown mandate (ie, the end of May 2021), recruitment letters with survey links were electronically shared with the principals at the participating schools, who then distributed the links to the students. Students who were interested in participating in the research could click the links in the recruitment letters to access the online surveys. At the very beginning of the surveys, students were informed of the purpose of the research, privacy and confidentiality, and their rights to discontinue participation at any time. The online questionnaires contained items related to conscientiousness, prior online learning experience, achievement emotions and academic procrastination along with demographic information. Surveys were closed at the end of June 2021. Data were then downloaded from the online platform and de-identified before the main analysis.

Measures

Conscientiousness

Nine 5-point Likert scale items (1 = Strongly Disagree, 5 = Strongly Agree) from John et al.'s (1991) Big Five inventory were adopted to assess students' conscientiousness. The scale is composed of 5 positively and 4 negatively worded items. The result of confirmatory factor analysis did not support a one-factor solution; rather, a two-factor model, consisting of the proactive and inhibitive aspects of conscientiousness, fit the data better (please see supplementary online material). Therefore, we proceeded with the two-factor solution in the subsequent analyses. Sample items for the proactive aspect of conscientiousness are “I am someone who makes plans and follows through with them,” “I am someone who does a thorough job,” and “I am someone who perseveres until the task is finished.” Revelle's omega total[1] is 0.92. Sample items for the inhibitive aspect of conscientiousness are “I am someone who is easily distracted,” “I am someone who can be somewhat careless,” and “I am someone who tends to be lazy.” Revelle's omega total is 0.86.

Prior online learning experience

One seven-point Likert scale item (1 = Extremely Inexperienced, 7 = Extremely Experienced) adapted from Artino (2008b) was used to assess students' online learning experience prior to the pandemic. The item is “How experienced were you with online learning prior to the breakout of COVID-19?”

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Achievement emotions

Five subscales from the short version of the Achievement Emotions Questionnaire (AEQ-S) were adapted to measure students' class-related emotions in online learning environments during the pandemic (Bieleke et al., 2021). Enjoyment represents students' positive emotion, while anger, anxiety, hopelessness and boredom represent negative emotions. All the subscales were assessed by 4 five-point Likert scale items (1 = Strongly Disagree, 5 = Strongly Agree). A sample item for enjoyment is “I enjoy being in online learning classes.” Revelle's omega total is 0.94. A sample item for anger is “I wish I didn't have to attend online learning classes because it makes me angry.” Revelle's omega total is 0.96. A sample item for anxiety is “I get tense in class in online learning classes.” Revelle's omega total is 0.93. A sample item for hopelessness is “I feel hopeless in online learning classes.” Revelle's omega total is 0.97. A sample item for boredom is “I get bored in online learning classes.” Revelle's omega total is 0.95.

We chose to measure more negative emotions than positive emotions and selected enjoyment as the only aspect of positive emotions for the following reasons. First, the current study was situated in the context of COVID-19. As research has shown, students are more likely to experience negative emotions than positive emotions amid the rapid transition to online learning environments (Espino et al., 2021). Second, the recent development in theory on procrastination has pointed out the role of emotion as an important antecedent. Particularly, procrastination mainly stems from negative emotions experienced during the goa-directed process (Baumeister et al., 2007; Sirois, 2016; Sirois & Pychyl, 2013). Prior studies have also shown that academic procrastination is more strongly correlated with negative emotions than positive emotions in remote settings (Henderikx et al., 2019; Rahimi & Vallerand, 2021). Third, enjoyment, in particular, has been selected by many studies to explore its role in relation to students' self-regulatory processes in online learning environments (Artino & Jones, 2012; Henderikx et al., 2019; Touati, 2020; You & Kang, 2014). These studies have consistently shown that enjoyment is significantly correlated with students' adaptive and maladaptive self-regulation in online learning environments. Fourth, of few studies on emotion and academic procrastination in online learning environments, Rahimi and Vallerand (2021) did not find significant correlations between positive emotions and academic procrastination. Henderikx et al. (2019), in particular, did not find a significant correlation between enjoyment and academic procrastination in online learning environments. These findings seem to contradict the theory and research findings in face-to-face settings (Rahimi, 2019). Therefore, there is a need to further examine the association between enjoyment and academic procrastination in online learning environments.

Academic procrastination

Five 5-point Likert scale items (1 = Strongly Disagree, 5 = Strongly Agree) were adapted from Yockey's (2016) study to assess students' procrastination in online learning environments. A sample item is “I know I should work on coursework in online learning classes, but I just don't do it.” Revelle's omega total is 0.96.

Covariate

Age as a continuous variable was included as a covariate given that theoretical frameworks have posited that personal characteristics can serve as important antecedents of achieve-

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ment emotions and students' self-regulatory processes (Pekrun et al., 2007; Pintrich & Zusho, 2007).

All the survey items were originally worded in English. Since participants' first language was Chinese, the forward and backward translation was employed in this study (Yu et al., 2004). The process involved two phases (Cederfeldt et al., 2015). First, all the instruments were translated from English to traditional Chinese independently by the first and second author, whose primary language was Chinese and both of whom were proficient in English. The translated items were then compared. Any discrepancies were discussed and resolved. The second phase was a backward translation done by the first and second author, respectively. Each translated item was then compared with the source document and assessed for the equivalence by the third and fourth authors, who were native English speakers. Any discrepancies were discussed and resolved by all the authors at the end of this phase.

Analytic approaches

Structural equation modelling

To answer Hypotheses 1–5, structural equation modelling (SEM) was implemented. We adopted the two-step approach as suggested by Kline (2016). First, confirmatory factor models were built by allowing conscientiousness, prior online learning experience, achievement emotions, academic procrastination and age to be freely correlated with one another. In addition, no structural relations were specified. Second, structural models were built where academic procrastination was regressed on conscientiousness, prior online learning experience, achievement emotions while controlling for age. In addition, achievement emotions were regressed on conscientiousness and prior online learning experience while controlling for age. Following the modelling strategy by Pekrun et al. (2009) and Theobald et al. (2021), we built measurement and structural models separately for each achievement emotion to avoid multicollinearity. A statistical model was discerned acceptable if global goodness-of-fit measures met the following benchmarks: comparative fit index (CFI) ≥ 0.90, the Tucker-Lewis index (TLI) ≥ 0.90, room mean square error of approximation (RMSEA) ≤ 0.06 and the standardized root mean square residual (SRMR) ≤ 0.08 (Hu & Bentler, 1999). The local misfit was evaluated by assessing the magnitude of factor loadings.

To answer Hypotheses 6 and 7, bias-corrected bootstrapping confidence intervals (BCBCI) with 5000 resamples were inspected to assess the indirect effects of conscientiousness and prior online learning experience on academic procrastination while controlling for age (Preacher & Hayes, 2008). There was no missing value in the analytic sample. The Type = Complex command in *Mplus* 8.4 was used throughout the analysis to handle the clustered data structure given that students were from different schools. Statistical significance was evaluated at a p value of 0.05 using a two-tailed test. Figure 1 presents the statistical model.

RESULTS

Descriptive statistics and bivariate correlations

Table 1 presents descriptive statistics of conscientiousness, prior online learning experience, achievement emotions and academic procrastination. In general, students had higher levels of the proactive aspect of conscientiousness than the inhibitive aspect of conscientiousness. They also had moderate levels of prior online experience, achievement emotions and

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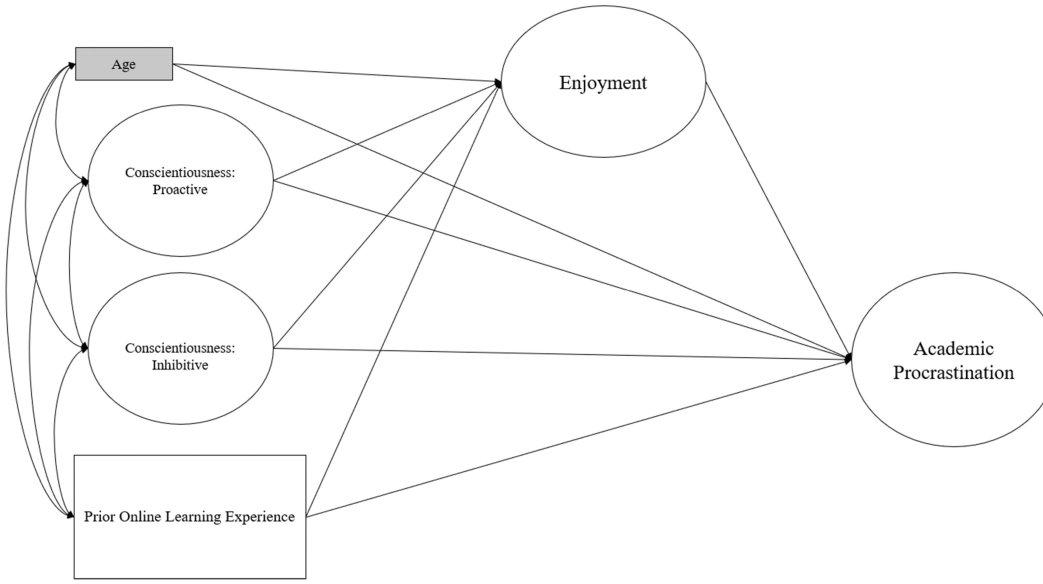

FIGURE 1 Statistical model.

academic procrastination. Among all the achievement emotions, students reported having the highest levels of enjoyment. Table 2 presents the results of the latent bivariate correlations. The proactive aspect of conscientiousness was significantly and positively correlated with the inhibitive aspect of conscientiousness, prior online learning experience, enjoyment and significantly and negatively correlated with anxiety, hopelessness, boredom and academic procrastination. The inhibitive aspect of conscientiousness was significantly and negatively correlated with anger, anxiety, hopelessness, boredom and academic procrastination. Prior online learning experience was significantly and positively correlated with enjoyment and negatively correlated with anger, hopelessness and boredom. Enjoyment was significantly and negatively correlated with negative emotions and academic procrastination. Negative emotions were significantly and positively correlated with one another and academic procrastination.

Structural relations

Table 3 presents the fit indexes for measurement and structural models. All the models demonstrated a satisfactory fit for the data with CFI and TLI greater than the cut-off values and RMSEA and SRMR below the cut-off values. Figure 2 presents the structural relations between conscientiousness, prior online learning experience, enjoyment and academic procrastination. Both aspects of conscientiousness and enjoyment significantly and negatively predicted academic procrastination. The proactive aspect of conscientiousness and prior online learning experience significantly and positively predicted enjoyment. Figure 3 presents the structural relations between conscientiousness, prior online learning experience, anger and academic procrastination. Both aspects of conscientiousness significantly and negatively predicted academic procrastination. Anger significantly and positively predicted academic procrastination. The inhibitive aspect of conscientiousness and prior online learning experience significantly and negatively predicted anger. Figure 4 presents the structural relations between conscientiousness, prior online learning experience, anxiety and

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TABLE 1 Descriptive statistics

Variables	*M*	*SD*	Min	Max	Skewness	Kurtosis
Conscientiousness: proactive	3.50	0.72	1.20	5.00	0.21	0.01
Conscientiousness: inhibitive	2.88	0.79	1.00	5.00	0.07	0.21
Experience	3.07	1.21	1.00	5.00	−0.20	−0.76
Enjoyment	3.11	1.00	1.00	5.00	0.03	−0.31
Anger	2.29	1.01	1.00	5.00	0.51	−0.18
Anxiety	2.33	0.95	1.00	5.00	0.38	−0.28
Hopelessness	2.08	0.99	1.00	5.00	0.65	−0.21
Boredom	2.73	1.06	1.00	5.00	0.10	−0.59
Academic procrastination	2.34	1.01	1.00	5.00	0.34	−0.56

TABLE 2 Latent bivariate correlations

	1	2	3	4	5	6	7	8	9
1. Conscientiousness: proactive	—								
2. Conscientiousness: inhibitive	0.36*	—							
3. Experience	0.32*	0.04	—						
4. Enjoyment	0.24*	−0.02	0.24*	—					
5. Anger	−0.07	−0.23*	−0.10*	−0.45*	—				
6. Anxiety	−0.11*	−0.27*	−0.09	−0.35*	0.82*	—			
7. Hopelessness	−0.12*	−0.26*	−0.14*	−0.30*	0.80*	0.89*	—		
8. Boredom	−0.12*	−0.30*	−0.10*	−0.44*	0.70*	0.72*	0.70*	—	
9. Academic procrastination	−0.24*	−0.39*	−0.05	−0.15*	0.49*	0.53*	0.61*	0.58*	—

*indicates $p < 0.05$; prior online learning experience is a manifest variable.

TABLE 3 Fit indexes for measurement and structural models

Model	χ^2	*df*	CFI	TLI	RMSEA	SRMR
Measurement						
Enjoyment	414.225*	157	0.959	0.950	0.047	0.041
Anger	398.884*	157	0.962	0.954	0.045	0.035
Anxiety	370.442*	157	0.959	0.951	0.043	0.034
Hopelessness	431.095*	157	0.958	0.949	0.048	0.037
Boredom	446.169*	157	0.951	0.941	0.050	0.039
Structural						
Enjoyment	414.226*	157	0.959	0.951	0.047	0.041
Anger	398.885*	157	0.962	0.954	0.045	0.035
Anxiety	370.421*	157	0.959	0.951	0.043	0.034
Hopelessness	431.094*	157	0.958	0.949	0.048	0.037
Boredom	446.168*	157	0.951	0.941	0.050	0.039

*refers to $p < 0.05$.

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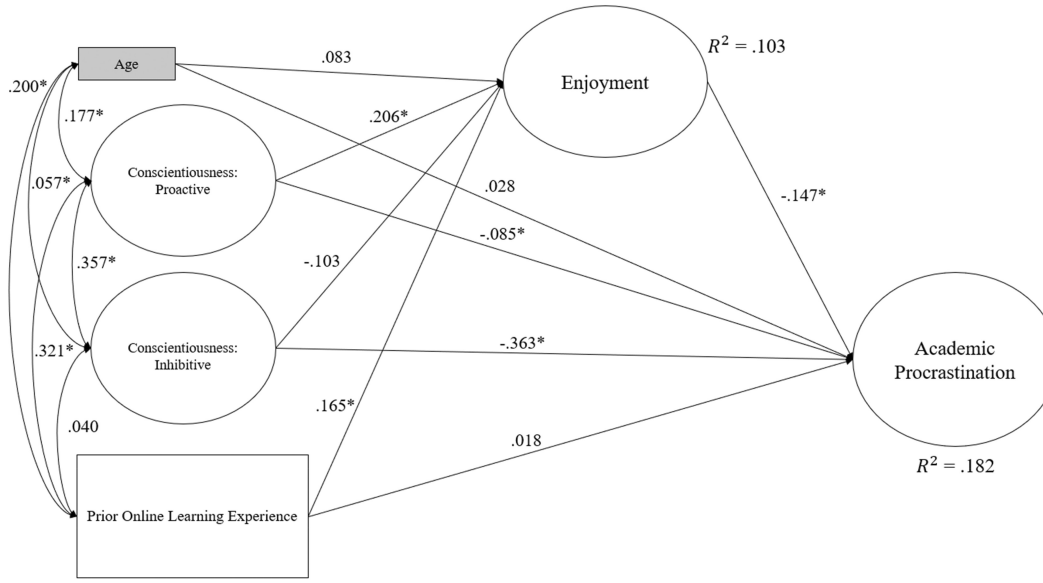

FIGURE 2 Structural model: Enjoyment. * refers to $p < 0.05$.

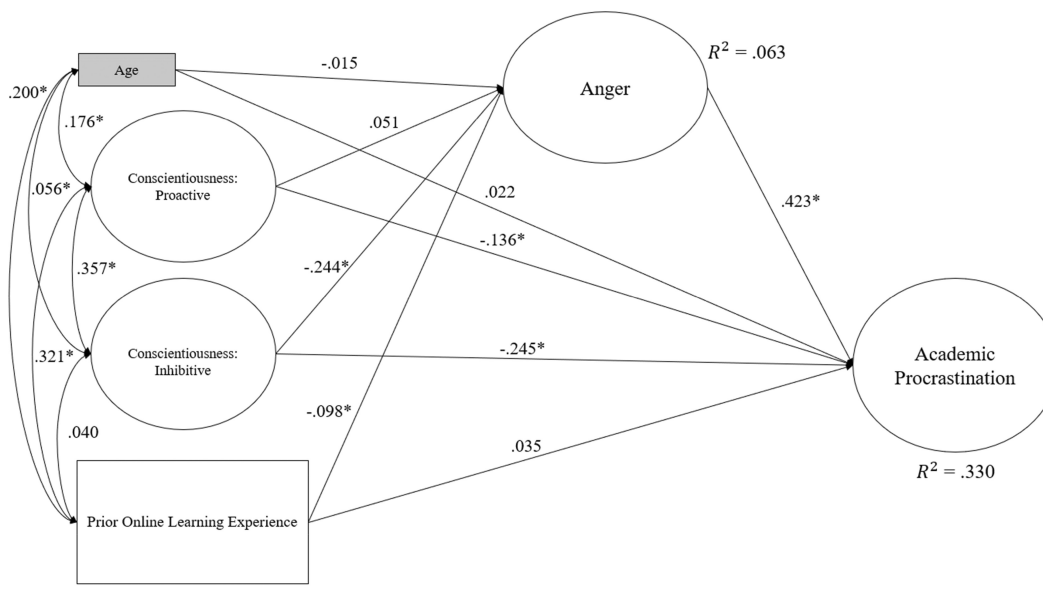

FIGURE 3 Structural model: Anger. * refers to $p < 0.05$.

academic procrastination. Both aspects of conscientiousness significantly and negatively predicted academic procrastination. Anxiety significantly and positively predicted academic procrastination. The inhibitive aspect of conscientiousness and prior online learning experience significantly and negatively predicted anxiety.

Figure 5 presents the structural relations between conscientiousness, prior online learning experience, hopelessness and academic procrastination. Both aspects of conscientiousness significantly and negatively predicted academic procrastination. Hopelessness significantly and positively predicted academic procrastination. The inhibitive aspect of conscientious-

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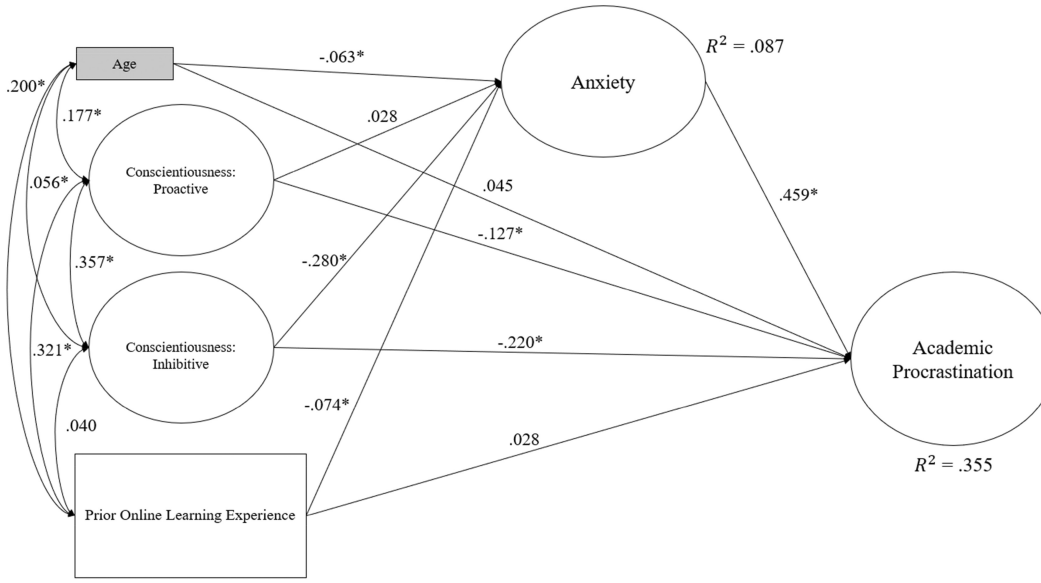

FIGURE 4 Structural model: Anxiety. * refers to $p < 0.05$.

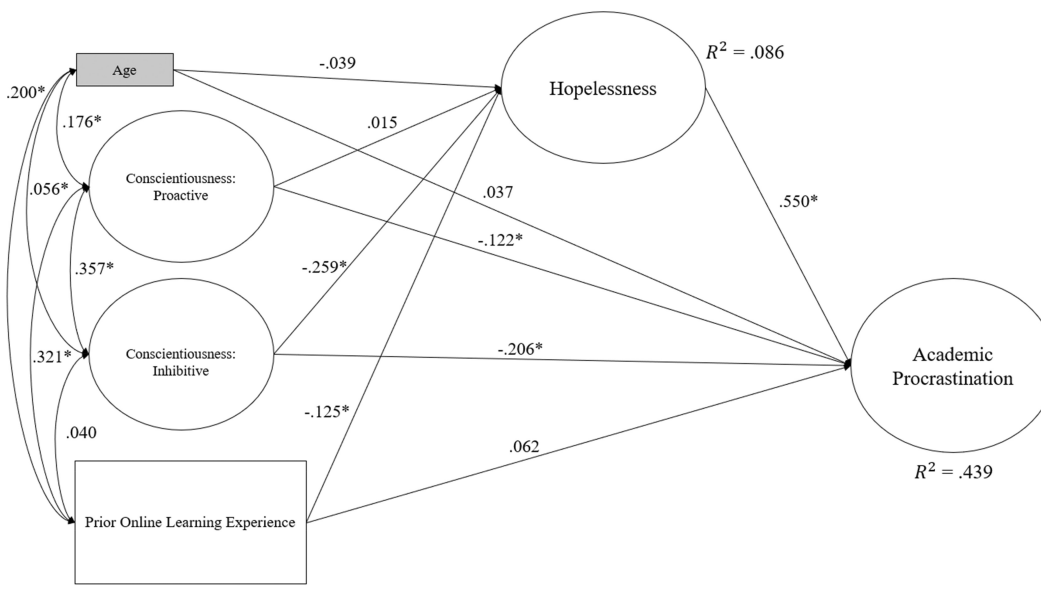

FIGURE 5 Structural model: Hopelessness. * refers to $p < 0.05$.

ness and prior online learning experience significantly and negatively predicted hopelessness. Figure 6 presents the structural relations between conscientiousness, prior online learning experience, boredom and academic procrastination. Both aspects of conscientiousness significantly and negatively predicted academic procrastination. Boredom significantly and positively predicted academic procrastination. The inhibitive aspect of conscientiousness and prior online learning experience significantly and negatively predicted boredom.

Table 4 presents the indirect effects of conscientiousness, prior online learning experience on academic procrastination. The proactive aspect of conscientiousness was significantly

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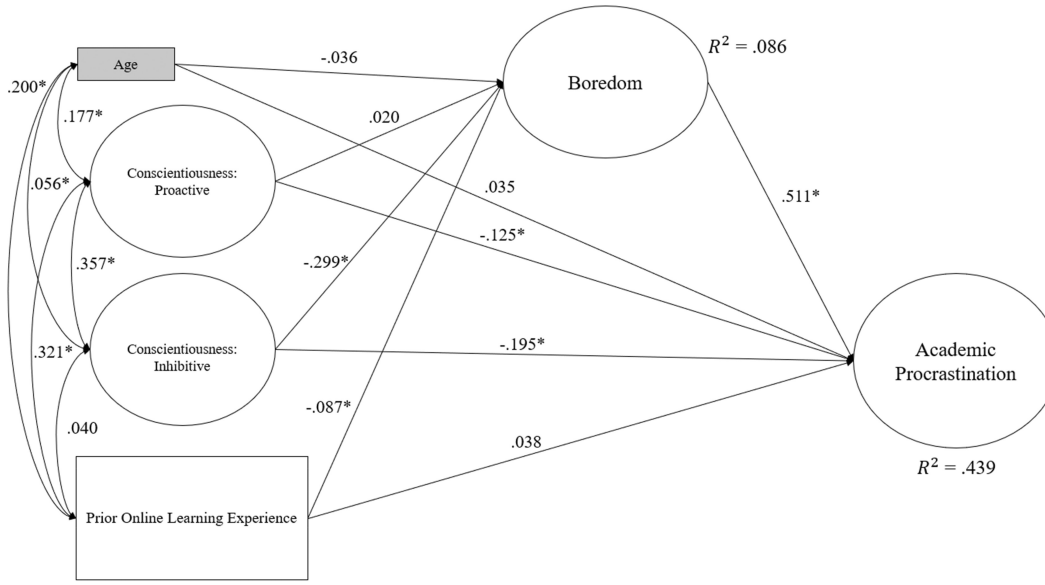

FIGURE 6 Structural model: Boredom. * refers to $p < 0.05$.

and negatively related to academic procrastination through the pathway of enjoyment. The inhibitive aspect of conscientiousness was significantly and negatively related to academic procrastination through the pathways of anger, anxiety, hopelessness and boredom. Prior online learning experience was significantly and negatively related to academic procrastination through the pathways of enjoyment, anger and hopelessness.

DISCUSSION

The goal of this study was to examine the structural relations between conscientiousness, prior online learning experience, achievement emotions and academic procrastination in online learning environments. The findings of the present study revealed several important insights into students' procrastination in online learning environments. Regarding Hypothesis 1 that positive emotions do not relate to academic procrastination, while negative emotions positively relate to academic procrastination in online learning environments, the current results partially support the hypothesis. In terms of the structural relations, enjoyment was significantly and negatively related to academic procrastination. Significant and positive relations were found for negative emotions. In particular, negative emotions were more strongly related to academic procrastination in online learning environments than positive emotions. These findings advance the existing literature in four ways. First, both Henderikx et al. (2019) and Rahimi and Vallerand (2021) did not find significant correlations between positive emotions and academic procrastination in online learning environments. Although these early findings contradict the current ones, the common thread among these studies is that academic procrastination is more closely associated with negative emotions than positive ones. Second, the cumulative findings of these studies seem to support the proposition that negative affective states precede procrastination. That is, when individuals experience negative emotions during the goal-directed process, they are more likely to succumb to procrastination to avoid task engagement and to seek immediate mood repair (Sirois, 2016). However, since these studies are relational in nature, the stated causal impact needs to be verified

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TABLE 4 Indirect effects of conscientiousness and prior online learning experience on academic procrastination

Paths	Point estimate (standardized)	95% CI Lower bound	95% CI Upper bound
Conscientiousness: proactive > enjoyment > procrastination	−0.030*	−0.063	−0.006
Conscientiousness: inhibitive > enjoyment > procrastination	0.015	−0.001	0.072
Experience > enjoyment > procrastination	−0.024*	−0.061	−0.002
Conscientiousness: proactive > anger > procrastination	0.021	−0.014	0.052
Conscientiousness: inhibitive > anger > procrastination	−0.103*	−0.150	−0.043
Experience > anger > procrastination	−0.042*	−0.064	−0.014
Conscientiousness: proactive > anxiety > procrastination	0.013	−0.043	0.046
Conscientiousness: inhibitive > anxiety > procrastination	−0.129*	−0.191	−0.054
Experience > anxiety > procrastination	−0.034	−0.064	0.000
Conscientiousness: proactive > hopelessness > procrastination	0.008	−0.050	0.043
Conscientiousness: inhibitive > hopelessness > procrastination	−0.142*	−0.195	−0.061
Experience > hopelessness > procrastination	−0.069*	−0.107	−0.034
Conscientiousness: proactive > boredom > procrastination	0.010	−0.048	0.062
Conscientiousness: inhibitive > boredom > procrastination	−0.153*	−0.201	−0.063
Experience > boredom > procrastination	−0.044	−0.076	0.011

*refers to $p < 0.05$.

by future experimental research. Third, as described in the literature review, self-regulation includes adaptive and maladaptive aspects (Baumeister et al., 1994). Academic procrastination is considered a type of maladaptive self-regulation (Balkis & Duru, 2016; Baumeister et al., 2007). According to the control-value theory, achievement emotions are hypothesized to be important antecedents of individuals' self-regulation (Pekrun et al., 2002, 2007). Previous research has presented evidence to support the relation between achievement emotions and students' adaptive self-regulation in online learning environments (Artino & Jones, 2012; You & Kang, 2014). Evidence on the relation between achievement emotions and maladaptive self-regulation such as academic procrastination in online learning environments was limited. The current study further displays that achievement emotions are also associated with students' maladaptive self-regulation (ie, academic procrastination) in online learning environments. Taken together, the cumulative findings show that students' emotional experiences are associated with both adaptive and maladaptive self-regulation in online learning environments. Last, meta-analyses have consistently shown that among all the factors, conscientiousness has the strongest correlation with procrastination. However, the current study shows that in online learning environments, negative emotions are even more strongly correlated with academic procrastination than conscientiousness is.

Regarding Hypothesis 2 that conscientiousness negatively relates to academic procrastination in online learning environments, the current results support the hypothesis. In all the structural models, both aspects of conscientiousness were negatively related to academic procrastination in online learning environments, which supports the theory that the proactive and inhibitive aspects of conscientiousness represent the same valence (Roberts et al., 2005). However, the inhibitive aspect of conscientiousness had a stronger association with academic procrastination than the proactive aspect of it. This finding suggests that students' unnecessary postponement in online classes is more closely related to their

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tendency to control their thoughts, feelings and behaviours than their striving for achievement and commitment to coursework. Also, although both aspects of conscientiousness were negatively correlated with academic procrastination, the magnitude was smaller than what has been reported by meta-analyses in the context of general settings (Steel, 2007; van Eerde, 2003).

Regarding Hypothesis 3 that conscientiousness positively relates to positive emotions and negatively relates to negative emotions in online learning environments, the current results partially support the hypothesis. In terms of the structural relations, the proactive aspect of conscientiousness was significantly and positively related to enjoyment but not significantly related to negative emotions. On the other hand, the inhibitive aspect of conscientiousness was significantly and negatively related to negative emotions but not significantly related to enjoyment. Although the current findings seem to be different from what has been found previously in general life settings (Carver et al., 2010; Fayard, 2012; Fayard et al., 2012; Watson & Clark, 1992), the relational patterns, in general, are similar. That is, proactive conscientiousness is strongly related to positive emotions, while inhibitive consciousness is more strongly related to negative emotions, which aligns with the findings of bivariate correlations and structural models in the current study. In sum, the proactive and inhibitive aspects of conscientiousness have differential relations with achievement emotions in online learning environments. Therefore, differentiating these two aspects of conscientiousness is needed.

Regarding Hypothesis 4 that prior online learning experience negatively relates to academic procrastination in online learning environments, the current results do not support the hypothesis. Prior online learning experience was not significantly related to academic procrastination in all the structural models. In addition, prior online learning experience was not significantly correlated with academic procrastination (Table 2). Taken together, these findings suggest that although prior online learning experience has a direction association with students' adaptive self-regulation (Li, 2019; Wang et al., 2013), it has no direct associations with academic procrastination in online learning environments.

Regarding Hypothesis 5 that prior online learning experience positively relates to positive emotions and negatively relates to negative emotions in online learning environments, the current results support the hypothesis. In terms of the structural relations, prior online learning experience was significantly and positively related to enjoyment and negatively related to anger, anxiety, hopelessness and boredom. Specifically, prior online learning experience had a stronger association with enjoyment than with negative emotions. Therefore, students with more experience in online learning are much more likely to experience positive emotions and less likely to experience negative emotions in online learning environments. Previous research has focused on the relation between prior online learning experience and negative emotions (Abdous, 2019; Artino, 2008a). The current results advance the literature by demonstrating the importance of prior online learning experiences in relation to diverse achievement emotions in online learning environments.

Regarding Hypotheses 6 and 7 that conscientiousness and prior online learning experience indirectly relate to academic procrastination through the pathways of achievement emotions in online learning environments, the current results support both hypotheses and the theoretical model proposed by Pekrun (2006). The findings on the indirect effects reveal the mechanism underlying the relation between conscientiousness, prior online learning experience and academic procrastination in online learning environments. As Table 4 shows, apart from the direct effect of the proactive aspect of conscientiousness on academic procrastination, this aspect of conscientiousness also had a significant indirect relation with academic procrastination through the mechanism of enjoyment. However, the indirect effects of the proactive aspect of conscientiousness on academic procrastination through the pathways of negative emotions were not statistically significant. The difference is more likely to be attributed to the fact that proactive conscientiousness is more closely related to positive

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emotions rather than negative emotions (Carver et al., 2010; Costantini & Perugini, 2016; Fayard, 2012).

On the other hand, apart from the direct effect of the inhibitive aspect of conscientiousness on academic procrastination, this aspect of conscientiousness also had significant indirect relations with academic procrastination through the mechanism of negative emotions. However, the indirect effect of the inhibitive aspect of conscientiousness on academic procrastination through the pathway of enjoyment was not statistically significant. Similarly, the difference is more likely to be attributed to the fact that inhibitive conscientiousness is more closely related to negative emotions rather than positive emotions (Fayard et al., 2012).

Taken together, the current findings suggest that one of the possible reasons students with high proactive conscientiousness are less likely to procrastinate in online learning environments is that they are more likely to experience enjoyment, which is related to their low tendency to postpone their coursework until the last minute. In a similar vein, one of the possible reasons students with high inhibitive conscientiousness are less likely to procrastinate in online learning environments is that they are less likely to experience negative emotions, which is related to their low tendency to delay unnecessarily.

Although prior online learning experience had no direct associations with academic procrastination as found in the present study, experienced online learners were still less likely to put off their coursework in online classes owing to their higher likelihood to experience enjoyment and lower likelihood to experience negative emotions than novice peers (Abdous, 2019; Artino, 2008a). In sum, the current findings identify important students' characteristics that will be associated with procrastination in online classes and explain why these students may engage in such maladaptive self-regulation.

Limitations and future directions

There are seven limitations associated with the current study. First, the current study is based on observational data and relational in nature. Therefore, no causal inferences can be drawn from the findings. Experimental and longitudinal research is needed to determine the causal effects of conscientiousness and prior online learning experience on achievement emotions and academic procrastination in online learning environments. Second, as Pekrun (2006) suggested, the general mechanisms of achievement emotions are bound to be similar across individuals; however, patterns and intensity of achievement emotions can be different between cultures. The current study is contextualized, specifically in Asia. The predictive patterns displayed in structural models could be different in other cultural contexts. This will be an important avenue for future research to confirm this hypothesis. Third, the current study was conducted during the COVID-19 pandemic. As Hodges et al. (2020) pointed out, learning experiences amid the rapid transition to remote teaching could be different from those during the normal type of online education. Therefore, there could be discrepancies between the current findings and the ones found in the normal school context. Replication studies are warranted. Fourth, the current study focused on the quantitative aspect of prior online learning experience. The qualitative aspect of students' prior experience in online courses such as enjoyable or boring was not examined. It is possible that the qualitative aspect of prior online learning experience could have differential relations with students' subsequent achievement emotions and academic procrastination in online learning environments. Researchers are encouraged to include this aspect in their future studies. Fifth, although there are some known advantages of internet surveys such as low unit cost, high spend of returns and protection of privacy, the quality of answers provided could be in question if no interviewers are present (Fowler, 2014). Researchers are encouraged to employ multiple modes of data collection to increase the validity of collected data. Sixth, since the

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current study is based on a convenience sample, there is no guarantee that the measurement and structural models can be generalized to the whole population. More studies are needed to examine if the current findings can be replicated in different contexts. If possible, probability sampling procedures can be used to increase the precision of sample estimates and the external validity of the findings (Fowler, 2014). Last, the current study focused on testing mediation models where conscientiousness and prior online learning experience were related to academic procrastination through the pathways of achievement emotions. Therefore, interaction between exogenous variables were not considered. It is possible that the relation between conscientiousness and academic procrastination is conditional on prior online learning experience. Researchers are encouraged to explore this direction in the future.

Implications and conclusions

The current study offers several theoretical implications. First, the conceptualization of conscientiousness as two distinct aspects is supported (Costa et al., 1991; Roberts et al., 2005). Although proactive and inhibitive conscientiousness are correlated with one another, their relations with emotions and procrastination are somewhat different. Proactive conscientiousness is more closely related to positive emotions, while inhibitive conscientiousness is more closely related to negative emotions. In addition, inhibitive conscientiousness is more strongly related to academic procrastination than proactive consciousnesses is. The findings suggest that proactive conscientiousness acts as the activator of the positive valence of achievement emotions and adaptive self-regulation, while inhibitive conscientiousness acts as the activator of the negative valence of achievement emotions and maladaptive self-regulation in online learning environments. Hence, there is a need to differentiate these two aspects of conscientiousness in theory and future research.

Second, as suggested by the principles of relative universality (Pekrun, 2018), the mechanisms underlying achievement emotion are universal across individuals. However, the intensity, frequency and patterns of emotions could differ by personal characteristics. The current findings support this proposition as conscientiousness and prior online learning experience are significantly related to achievement emotions in online learning environments. The current study advances the theory by identifying specific personal characteristics that need to be considered when it comes to students' emotional experiences in online learning environments.

Third, according to the general learning model proposed by Pintrich and Zusho (2007), personal characteristics serve as immediate antecedents of student' self-regulation. However, the current findings show that only conscientiousness is directly related to students' academic procrastination in online learning environments. Prior online learning experience does not have a direct effect on academic procrastination. Therefore, whether personal characteristics are directly related to self-regulation may depend on the nature of specific personal factors. More likely, personal characteristics are related to students' self-regulation through other psychological mechanisms such as achievement emotions as will be discussed later.

Fourth, achievement emotion as an important antecedent of students' self-regulation as suggested by the control-value theory (Pekrun, 2006; Pekrun et al., 2002) is supported. In particular, achievement emotions are related to not only students' adaptive self-regulation (Artino & Jones, 2012; Kim et al., 2014; You & Kang, 2014) but also their maladaptive self-regulation (ie, academic procrastination) in online learning environments. Further, previous findings along with the current ones suggest that positive emotions are more strongly related to adaptive self-regulation, while negative emotions are more strongly related to maladaptive self-regulation. However, how a specific valence of achievement emotions is

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related to a specific valence of self-regulation has not been discussed much in theory. We believe this is a meaningful aspect of theory to explore in future research.

Fifth, the current findings also support the role of emotion as a major antecedent of procrastination as suggested by the recent development in the theory on procrastination (Sirois, 2016; Sirois & Pychyl, 2013). Conscientiousness has been demonstrated as the strongest factor associated with procrastination in conventional school settings (Steel, 2007). However, the current study shows that negative emotions are even more strongly associated with academic procrastination than either aspect of conscientiousness is in online learning environments. Therefore, when students' experience negative emotions during the goal-directed process, they are more likely to resort to procrastination as a counterproductive approach to avoid task engagement and negative affective states. Although conscientiousness, motivation and time management has been highlighted as the major contributors associated with procrastination in some of the major theoretical frameworks (Schraw et al., 2007; Steel, 2007; Wolters, 2003), the role of emotion is often overlooked and should be integrated into future theory building.

Sixth, as hypothesized by the control-value theory (Pekrun, 2006), achievement emotions mediate the relation between personal characteristics and self-regulation. The current findings support this proposition. Specifically, conscientiousness and prior online learning experience are indirectly related to academic procrastination in online learning environments through different mechanisms. Proactive conscientiousness is indirectly related to procrastination mainly through enjoyment, while inhibitive conscientiousness is indirectly related to procrastination mainly through negative emotions. Prior online learning experience are indirectly related to procrastination through both enjoyment and negative emotions. These findings suggest that when a global construct is composed of different subdomains, each subdomain may be indirectly related to self-regulation through the pathways of different valences of achievement emotions.

In terms of practical implications, the current study suggests that conscientiousness and online learning experience are reasonable targets for intervention in experimental research on reducing procrastination in online learning environments. Roberts et al. (2017) offered theoretical perspectives on how to change conscientiousness. Scholars who aim to target conscientiousness in future experimental studies are encouraged to consult their work. However, there is no established approach to enhance online learning experience in the existing literature. We encourage scholars to explore viable ways to manipulate students' perceptions of online learning experience within a reasonable period of time. Instructors are suggested to pay attention to students with low conscientiousness and novice online learners. These students are more likely to engage in unnecessary delay in online classes.

In conclusion, the goal of the current study was to examine the structural relations between conscientiousness, prior online learning experience, achievement emotions and academic procrastination in online learning environments. Students with lower conscientiousness and fewer prior online learning experience are more likely to procrastinate in online learning environments owing to their lower likelihood to experience enjoyment and higher likelihood of experience negative emotions in coursework. More research is needed to verify the findings reported here and examine academic procrastination in self-paced online learning environments.

FUNDING INFORMATION

None.

CONFLICT OF INTEREST

There is no potential conflict of interest in this study.

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DATA AVAILABILITY STATEMENT

The data used in this study can be accessed by sending request emails to the corresponding author.

ETHICS STATEMENT

Participation in this study was completely voluntary. All the personally identifiable information was de-identified upon being collected. All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards.

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ENDNOTE

[1] Revelle's omega total is an index of scale reliability alternative to coefficient alpha (α). Coefficient alpha assumes the condition of tau equivalence (ie, indicators of a given factor have equal factor loadings). However, this condition is seldom met in practice (including the current study) and, hence, alpha often does not provide consistent information about scale reliability (McNeish, 2018). Coefficient omega does not make such assumption and overcomes the weakness of alpha by accounting for the conditions of unequal factor loadings and correlated measurement errors (Brown, 2015). A cut-off reliability coefficient ≥0.70 is considered acceptable (Nunnally & Bernstein, 1994).

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